You saw the following context-free grammars in class on Wednesday; in each example, the grammar itself is on the left; the explanation for each non-terminal is on the right.

- Properly nested strings of parentheses.

$$S \to \varepsilon \mid S\,\texttt{(}\,S\,\texttt{)} \qquad \text{properly nested parentheses}$$

  Here is a different grammar for the same language:

$$S \to \varepsilon \mid \texttt{(}\,S\,\texttt{)} \mid SS \qquad \text{properly nested parentheses}$$

- $\{\texttt{0}^m\texttt{1}^n \mid m \neq n\}$. This is the set of all binary strings composed of some number of $\texttt{0}$s followed by a *different* number of $\texttt{1}$s.

$$\begin{aligned} S &\to A \mid B & \{\texttt{0}^m\texttt{1}^n \mid m \neq n\} \\ A &\to \texttt{0}A \mid \texttt{0}C & \{\texttt{0}^m\texttt{1}^n \mid m > n\} \\ B &\to B\texttt{1} \mid C\texttt{1} & \{\texttt{0}^m\texttt{1}^n \mid m < n\} \\ C &\to \varepsilon \mid \texttt{0}C\texttt{1} & \{\texttt{0}^m\texttt{1}^n \mid m = n\} \end{aligned}$$

Give context-free grammars for each of the following languages over the alphabet $\Sigma = \{\texttt{0}, \texttt{1}\}$. For each grammar, describe the language for each non-terminal, either in English or using mathematical notation, as in the examples above. We probably won't finish all of these during the lab session.

1. All palindromes in Σ^*

2. All palindromes in Σ^* that contain an even number of $\texttt{1}$s

3. All palindromes in Σ^* that end with $\texttt{1}$

4. All palindromes in Σ^* whose length is divisible by 3

5. All palindromes in Σ^* that do not contain the substring $\texttt{00}$

Harder problems to work on later:

6. $\{0^{2n}1^n \mid n \geq 0\}$

7. $\{0^m 1^n \mid m \neq 2n\}$

   [Hint: If $m \neq 2n$, then either $m < 2n$ or $m > 2n$. Extend the previous grammar, but pay attention to parity. This language contains the string 01.]

8. $\{0, 1\}^* \setminus \{0^{2n}1^n \mid n \geq 0\}$

   [Hint: Extend the previous grammar. What's missing?]

9. $\left\{w \in \{0, 1\}^* \;\middle|\; \#(0, w) = 2 \cdot \#(1, w)\right\}$ — Binary strings where the number of 0s is exactly twice the number of 1s.

*10. $\{0, 1\}^* \setminus \{ww \mid w \in \{0, 1\}^*\}$.

    *[Anti-hint: The language $\{ww \mid w \in 0, 1^*\}$ is **not** context-free. Thus, the complement of a context-free language is not necessarily context-free!]*